

Project Overview: Router-on-a-Stick Inter-VLAN Routing

Objective: To design and implement a scalable, secure, and segmented network infrastructure using Cisco Packet Tracer. This project demonstrates the practical application of inter-departmental communication through Router-on-a-Stick routing, ensuring efficient data segregation and secure traffic management across 4 distinct departments (IT, HR, Finance, and CS).

Key Implementations:

- **Network Segmentation:** Configured 4 VLANs (10, 20, 30, 40) to isolate departmental traffic.
- **Routing:** Implemented Router-on-a-Stick using sub-interfaces and 802.1Q encapsulation.
- **Security:** Enforced security best practices by configuring a dedicated Native VLAN (VLAN 99) and restricting trunk ports with Allowed VLANs.
- **Connectivity:** Validated end-to-end communication via ICMP verification across the entire topology.

IP Addressing Table

Department	VLAN	Network ID	Gateway (Router Sub-interface)	Range of Usable IPs
HR	10	192.168.10.0	192.168.10.1	192.168.10.2 - 192.168.10.254
IT	20	192.168.20.0	192.168.20.1	192.168.20.2 - 192.168.20.254
Finance	30	192.168.30.0	192.168.30.1	192.168.30.2 - 192.168.30.254

Department	VLAN	Network ID	Gateway (Router Sub-interface)	Range of Usable IPs
cs	40	192.168.40.0	192.168.40.1	192.168.40.2 - 192.168.40.254

Switch-1 configuration

enable

configure terminal

hostname S1

vlan 10

name HR

vlan 20

name IT

vlan 30

name FINANCE

vlan 40

name CS

vlan 99

name NATIVE

exit

interface range FastEthernet0/1-2

switchport mode access

switchport access vlan 10

exit

```
interface range FastEthernet0/3-4
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

```
interface range FastEthernet0/5-6
```

```
switchport mode access
```

```
switchport access vlan 30
```

```
exit
```

```
interface range FastEthernet0/7-8
```

```
switchport mode access
```

```
switchport access vlan 40
```

```
exit
```

Trunk to S2

```
interface FastEthernet0/23
```

```
switchport trunk encapsulation dot1q
```

```
switchport mode trunk
```

```
switchport trunk native vlan 99
```

```
exit
```

Trunk to R1

```
interface FastEthernet0/24
```

```
switchport mode trunk
```

```
switchport trunk native vlan 99
```

```
exit
```

end

write memory

Switch S2

enable

configure terminal

hostname S2

vlan 10

name HR

vlan 20

name IT

vlan 30

name FINANCE

vlan 40

name CS

vlan 99

name NATIVE

exit

interface range FastEthernet0/1-2

switchport mode access

switchport access vlan 10

exit

interface range FastEthernet0/3-4

switchport mode access

switchport access vlan 40

exit

interface range FastEthernet0/5-6

```
switchport mode access
```

```
switchport access vlan 30
```

```
exit
```

```
interface range FastEthernet0/7-8
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

Trunk

```
interface FastEthernet0/24
```

```
switchport mode trunk
```

```
switchport trunk native vlan 99
```

```
exit
```

```
end
```

```
write memory
```

ROUTER

```
enable
```

```
configure terminal
```

```
hostname R1
```

```
interface GigabitEthernet0/0
```

```
no shutdown
```

```
exit
```

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 exit
```

```
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 exit
```

```
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 exit
```

```
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 exit
```

```
interface GigabitEthernet0/0.99
 encapsulation dot1Q 99 native
 ip address 192.168.99.1 255.255.255.0
 exit
end
write memory
```

Verification commands

1. VLAN Verification (Switch)
Switch# show vlan brief
2. Trunk Verification
Switch# show interfaces trunk
3. Interface Status Check
Switch# show ip interface brief
Router# show ip interface brief
4. Router Subinterface Verification
Router# show running-config
OR more specific:
Router# show interfaces g0/0.10
Router# show interfaces g0/0.20
Router# show interfaces g0/0.30
Router# show interfaces g0/0.40
5. MAC Address Table (Switch)
Switch# show mac address-table
6. ARP Table (Router)
Router# show ip arp
7. Ping Test (End-to-End Connectivity)
ping 192.168.10.1
ping 192.168.20.1
ping 192.168.30.1
ping 192.168.40.1
8. Final Router Check
Router# show ip route